

Ex. No:6:-

Use Sleuth Kit to analyze digital evidence

Date:-

Aim:-

To use The Sleuth Kit command-line tools to analyze a disk image and extract digital evidence.

Description:-

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 - 1. 4Dell Latitude CPi.E01
 - 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command: `arduino fsstat [image file] > filesystem_info.txt`
- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

`arduino mmls [image file] > partitions.txt` ○ This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command: `arduino`

`fls -r [image file] > file_list.txt` ○ This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use `icat`:

`css icat [image file] [inode number] > [output file]` ○ Replace [inode number] with the inode of the file you want to recover, which you can find from the `fls` output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command: `css`

`istat [image file] [inode number] > metadata_info.txt` ○ This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using `fls`: `css fls -m / -r [image file] > body.txt` ○

Then create the timeline: `css`

`mactime -b body.txt > timeline.txt` ○ The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. **Compile the Data:** ○ Gather all the output files (e.g., `filesystem_info.txt`, `partitions.txt`, `file_list.txt`, `metadata_info.txt`, `timeline.txt`).
2. **Analyze and Document:**
 - Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures. By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

SCREENSHOTS :

```
C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
Missing image name
Usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
-t: display type only
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-p pooltype: Pool container type (use '-p list' for supported types)
-B pool volume block (Starting block for pool volumes only)
-v: verbose output to stderr
-V: Print version
-k password: Decryption password for encrypted volumes

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>dir
"C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp"
Volume in drive C is OS
Volume Serial Number is 462C-EA9E

Directory of C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp

27-08-2026  00:51      <DIR>          .
27-08-2026  00:50      <DIR>          ..
26-08-2026  22:24      671,094,597  4Dell Latitude CPi.E01
26-08-2026  22:22      419,084,951  4Dell Latitude CPi.E02
                2 File(s)  1,090,479,548 bytes
                2 Dir(s)  70,902,882,304 bytes free

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
"C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\4Dell Latitude CPi.E01" > "C:\Users\UDAY
KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"
Cannot determine file system type

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\User
\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users
\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\4Dell Latitude CPi.E01" > "C:\Users\UDAY KUMAR
REDDY\OneDrive\Desktop\sleuth kit exp\partitions.txt"

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Use
rs\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\partitions.txt"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot   Start      End      Length   Description
000:  Meta   0000000000  0000000000  0000000001  Primary Table (#0)
001:  000:000  0000000062  0009510479  0009510417  Unallocated
002:  000:000  0009510480  0009514979  0000004501  NTFS / exFAT (0x07)
003:  000:000  0009514980  0009514259  0000009780  Unallocated

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -r "C:
\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\4Dell Latitude CPi.E01" > "C:\Users\UDAY
KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"
Cannot determine file system type

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type
"C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\file_list.txt"
```

```

++ r/r 3923-128-4: cga40852.fon
++ r/r 3470-128-3: ega40850.fon
++ r/r 3425-128-3: mvboli.ttf
++ r/r 3412-128-3: smalle.fon
++ r/r 3455-128-3: timesbd.ttf
++ r/r 4000-128-4: vga869.fon
++ r/r 3965-128-4: vgaf1257.fon
++ r/r 3403-128-3: vgafix.fon
++ r/r 3936-128-4: vgafixe.fon
++ r/r 4040-128-4: vgafixr.fon
++ r/r 4067-128-4: vgafixt.fon
++ r/r 4041-128-4: vgasysr.fon
++ r/r 4068-128-4: vgasyst.fon
++ r/r 5028-128-4: Gone Fishing.bmp
++ r/r 5029-128-4: Greenstone.bmp
+ d/d 9846-144-6: Installer
++ r/r 10179-128-3: ac704.msi
++ r/r 9860-128-3: d2a9f.msi
+ d/d 9861-144-1: {350C97B0-3D7C-4EE8-BAA9-80BCB3DD54227}
++ r/d 1030-144-6: places.exe
++ d/d 10180-128-6: {6C31E111-96BB-4ADC-9C81-E6D3EEDDD8D3}
++ r/r 10183-128-4: ARPPRODDUCTION.exe
++ r/r 10181-128-3: htmlgen.exe
++ r/r 10182-128-4: PowerCalc.exe
++ r/r 10185-128-3: Readme
++ r/r 10184-128-4: Timershot.exe
++ r/r 10186-128-3: TweakUI.exe
++ r/r 11772-128-3: iun6002.exe
++ r/r 11734-128-4: Look@LAN Setup Log.txt
++ r/r 3651-128-3: explorer.scf
++ r/r 3653-128-3: msdfmap.ini
+ r/r 339-128-3: REGLOCS.OLD
++ r/r 3680-128-1: setuperr.log
++ r/r 5464-128-1: vb.ini
++ r/r 3901-128-4: regopt.log
++ r/r 431-128-3: SchedLgU.Txt
++ r/- * 0: ~GLC0000.TMP
++ r/- * 0: ~GLH0000.TMP
++ r/- * 0: ~GLH0001.TMP
+ v/v 12305: $OrphanFiles
r/r 343-128-1 hiberfil.sys
r/r 228-128-1 pagefile.sys
C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\kit exp\4Dell Latitude CPi.E01" > "C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\file_list.txt"

```

```

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>

```

```

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type
C:\Users\UDAY KUMAR REDDY\OneDrive\Desktop\sleuth kit exp\4Dell Latitude CPi.E01"
MFT Entry Header Values:
Entry: 346 Sequence: 348
$LogFile Sequence Number: 28130412
Allocated Directory
Links: 2

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden
Owner ID: 0
Security ID: 296 (S-1-5-21-2000478354-688789844-1708537768-1003)
Created: 2004-08-20 04:34:05.300928000 (India Standard Time)
File Modified: 2004-08-20 03:54:35.240342400 (India Standard Time)
MFT Modified: 2004-08-20 04:34:06.753016000 (India Standard Time)
Accessed: 2004-08-20 20:47:59.434230400 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Directory
Name: TEMPLA--1
Parent MFT Entry: 344 Sequence: 346
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 04:34:05.300928000 (India Standard Time)
File Modified: 2004-08-20 04:34:05.300928000 (India Standard Time)
MFT Modified: 2004-08-20 04:34:05.300928000 (India Standard Time)
Accessed: 2004-08-20 04:34:05.300928000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Directory
Name: Template
Parent MFT Entry: 344 Sequence: 346
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 04:34:05.300928000 (India Standard Time)
File Modified: 2004-08-20 04:34:05.300928000 (India Standard Time)
MFT Modified: 2004-08-20 04:34:05.300928000 (India Standard Time)
Accessed: 2004-08-20 04:34:05.300928000 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-3) Name: N/A Resident size: 82
Type: $FILE_NAME (48-3) Name: N/A Resident size: 94
Type: $INDEX_ROOT (144-6) Name: $I30 Resident size: 56
Type: $INDEX_ALLOCATION (160-4) Name: $I30 Non-Resident size: 4096 init_size: 4096
3282303 2037304 2037305 2037306 2037307 2037308 2037309 2037310
Type: $BITMAP (176-5) Name: $I30 Resident size: 8
C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>

```

```

C:\Users\UDAY KUMAR REDDY\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Users\UDAY
KUMAR REDDY\OneDrive\Desktop\Desktop\sleuth kit exp\Greenstone_metadata.txt"
MFT Entry Header Values:
Entry: 5029          Sequence: 2
$LogFile Sequence Number: 12931864
Allocated File
Links: 2

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ( )
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.091435200 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: GREENS-1.BMP
Parent MFT Entry: 458 Sequence: 458
Allocated Size: 26624 Actual Size: 26582
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.081420800 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: Greenstone.bmp
Parent MFT Entry: 458 Sequence: 458
Allocated Size: 26624 Actual Size: 26582
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.081420800 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-6) Name: N/A Resident size: 90
Type: $FILE_NAME (48-5) Name: N/A Resident size: 94
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 124
Type: $DATA (128-1) Name: N/A Non-Resident size: 26582 init_size: 26582

```

Rubrics:-

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:-

The disk image was analyzed with mmls, fls, istat, icat, and mactime. The partitions and files were listed, a file's metadata and timestamps were extracted, and a timeline of file activity was generated.